

PLAN TO BUILD (PTB)

Automation Test Case Document

Module-wise catalogue of implemented Cypress end-to-end suites

Application	Plan to Build (PTB)
Environment	staging.plantobuild.online
Framework	Cypress 15.x (JavaScript, Page Object Model)
Modules covered	6
Automated test cases	24
Automated test steps	293
Document status	For review

1. Purpose and Scope

This document lists every automated test case currently implemented in the PTB Cypress regression suite, organised by application module and broken down into the individual steps that each suite executes. It is intended as a review artefact: each entry maps one-to-one to a test in the codebase, so coverage can be assessed and gaps agreed before the next automation cycle.

Only tests that exist and run today are listed. Cases that are written but currently disabled are called out explicitly in the notes under the relevant module, together with the reason.

2. Test Environment and Framework

Application under test	Plan to Build (PTB) - web application
Base URL	https://staging.plantobuild.online
Automation tool	Cypress 15.x
Language	JavaScript (ES modules)
Design pattern	Page Object Model - specs under cypress/e2e, page objects under cypress/pages, selectors centralised under cypress/support/selectors
Supporting libraries	cypress-real-events (real OS mouse and keyboard events for drag-and-drop and hover-gated controls), exceljs (XLSX read, edit and verification), cypress-mochawesome-reporter
Reporting	Mochawesome HTML report with charts and embedded screenshots, written to cypress/reports
Viewport	1920 x 1080
Test isolation	Disabled - each spec runs as one continuous, ordered user journey
Failure evidence	Screenshot captured automatically on failure
Test data	Users are read from a fixture; object names are timestamp-stamped per run so assertions are unambiguous and runs do not collide
Cleanup	Every suite deletes the schedules, structures, folders, teams and projects it creates, so the environment is left in its original state

3. Coverage Summary

24 automated test cases across 6 modules, comprising 293 individually asserted test steps.

Module	Case ID	Test Case / Suite	Steps
Project Creation	PC-01	Manual Project Creation - Advanced Settings	15
	PC-02	Import Project Creation - Full Flow (Project Info Sheet)	4
Project Creation subtotal		2 test cases	19
Teams & Permissions	TP-01	Project-Level Teams & Permissions	10
Teams & Permissions subtotal		1 test case	10
Coordination	COR-01	Coordination - Files (upload, versioning, share, rename, delete)	32
	COR-02	Coordination - ToDo Side Panel, Smart Filter & Export	9
	COR-03	Coordination Workflows - ToDo status flow	9
Coordination subtotal		3 test cases	50

Module	Case ID	Test Case / Suite	Steps
Schedule / Planning	SCH-01	Task Creation & Deletion	13
	SCH-02	Task Status Verification (single task)	10
	SCH-03	Parent & Child Task Status Verification	10
	SCH-04	Status roll-up with two child tasks	9
	SCH-05	Status and hierarchy with three child tasks	6
	SCH-06	Child-of-Child (grandchild) Task Status Verification	10
	SCH-07	Side Panel Task Editing	13
	SCH-08	Predecessor Links from the Side Panel	6
	SCH-09	Cash Flow - Booking / Invoicing, Nalco and Dashboard roll-up	11
	SCH-10	Planned vs Actual (snapshot baselining and comparison)	30
	SCH-11	Smart Filters	11
	SCH-12	Resources and Timesheet (multi-user)	12
	SCH-13	Share - Shareable Link (scope, settings and filters)	11
Schedule / Planning subtotal		13 test cases	152
Element Management System (EMS)	EMS-01	Building Models - Model	15
	EMS-02	Building Models - System	12
	EMS-03	Building Models - Drawing	11
	EMS-04	Building Models - Batch and QR Code	16
Element Management System (EMS) subtotal		4 test cases	54
Field Logs	FL-01	Form Template create, send, reopen, export & delete	8
Field Logs subtotal		1 test case	8
TOTAL		24 test cases	293

4. Module-wise Test Cases

Each step below corresponds to a single assertion block in the automation code. Steps run in sequence within a test case; a failure stops that case at the failing step.

4.1 Project Creation

2 test cases | 19 steps

Covers both routes for onboarding a new project into PTB: the Advanced (manual) wizard and the Import route driven by a Project Info Sheet (XLSX). The Import suite additionally cross-validates every value in the uploaded workbook against the rendered Project Info Sheet checklist.

PC-01 - Manual Project Creation - Advanced Settings

Spec file	cypress/e2e/PTB_ProjectCreation/ManualProjectCreation.cy.js
Suite name	Manual Project Creation - Advanced Settings
Executed as	Import user (Schueco India)
Pre-conditions	User logged in; company switched to Schueco India.
Step count	15

Step	Action	Expected Result
1	Switch the active company to Schueco India (re-authenticate when prompted).	Company context changes to Schueco India and the project dashboard loads.
2	Click New, then choose Advanced.	The multi-step Create Project wizard opens on the Project Type screen.
3	Keep Project Type = New Building (default) and click Next.	Wizard advances to the Basic Info screen.
4	Enter a unique Project Name and Project Number.	Both values are accepted; no duplicate-number validation error is raised.
5	Select Status = Active.	Status field shows Active.
6	Clear the Start Date and pick the current date.	Start Date is set to today.
7	Click Next.	Wizard advances to the Location screen.
8	Fill Address Line 1, City, State, Zip and Country.	All location fields are populated and accepted.
9	Click Next.	Wizard advances to the Sales screen.
10	On the Sales screen click Next.	Wizard advances to Contact Information.
11	On Contact Information click Next.	Wizard advances to the Folder Structure Template screen.
12	Select the first Folder Structure Template and click Next.	Template is selected; wizard advances to the Schedule Template screen.
13	Select the first Schedule Template and click Next.	Template is selected; wizard advances to the Teams screen.
14	Select All Teams and click Next.	Teams are assigned; the Review screen is displayed.
15	Click Create.	Project is created and the app redirects to /app/project/portal/<projectId>.

PC-02 - Import Project Creation - Full Flow (Project Info Sheet)

Spec file	cypress/e2e/PTB_ProjectCreation/ImportProjectCreation.cy.js
Suite name	Import Project Creation - Full Flow (TC01-TC08)
Executed as	Import PM user (Schueco India)
Pre-conditions	Viewport forced to 1920x1080 so the Checklist entry stays in the project top bar. Source workbook Central IKON_PIS_Latest_version.xlsx is read up-front through a Cypress node task.
Step count	4

Step	Action	Expected Result
1	TC01 - Import the Project Info Sheet: New > Import Project > upload the XLSX; keep Project Type = New Building; overwrite the Project Number with a unique value; fill the required Location fields; step through Sales and Contact Information; skip the Folder Structure and Schedule templates; select team "Team Sidhi Automation"; click Create. Then re-login and reopen the project by Project Number.	Project is created and the app redirects to /app/project/portal/<projectId>. The project ID is captured for later steps and the Checklist entry is present in the project top bar.
2	TC02 - Open the generated Project Info Sheet from the Checklist list.	The Project Info Sheet renders and the company block shows the expected registered company address.
3	TC03 - Parse the uploaded XLSX into lettered sections (A, B, C, D...), normalise the layout (Section A two-pair rows flattened; Sections B/C/D leading index column stripped), scrape every rendered PIS section table and compare row by row.	Every rendered section has headers and at least one row, and every non-empty (title, value) pair from the uploaded workbook is found in the matching Project Info Sheet row.
4	TC08 - From the projects list, open the All tab, search the Project Number, open the project card and delete the project.	The project is deleted and no longer returned by a search on its Project Number.

Note

TC04-TC07 (PM adds a checklist row, non-PM read-only access check, edited-XLSX re-upload, and row cleanup) are written and retained in the spec but currently disabled. The add/edit/delete interactions inside `cmacs-compact-table` are gated by real operating-system hover state that Cypress cannot simulate reliably; these four cases are covered by manual verification. Re-enabling them is a matter of uncommenting the block.

4.2 Teams & Permissions

1 test case | 10 steps

End-to-end validation that project-level team membership and per-object permissions are actually enforced. Permissions are configured as an administrator and then verified by logging in as the restricted user.

TP-01 - Project-Level Teams & Permissions

Spec file	cypress/e2e/PTB_TeamsAndPermissions/ProjectLevel_Permissions.cy.js
Suite name	Project-Level Teams & Permissions (MJ)
Executed as	Admin / PM, then non-PM member (User MJ)
Pre-conditions	Project "Automation Project 3" is open.
Step count	10

Step	Action	Expected Result
1	Navigate to Teams and create a new team named MJ.	The team MJ appears in the Teams left panel.
2	Drag and drop User MJ and Michael J from Contacts into the MJ team column.	Both contacts are added as members of the MJ team.
3	Go to the Coordination workspace and upload two folders (perm-folder-1 and perm-folder-2).	Both folders are uploaded and listed in Coordination.
4	Open Permissions for the MJ team and set document permissions: first document = Read Only, second document = Full Access.	Each permission is saved (inline save spinner clears) and the row reflects the selected access level.
5	Go to the Schedule workspace, create a schedule and add two tasks.	Schedule is created and two tasks exist in the grid.
6	Return to Permissions, open the Schedule tab for the MJ team, select the new schedule, then set task permissions: task 1 = No Access, task 2 = Limited Access.	Both task-level permissions are saved successfully.
7	Log out and log in as User MJ; open the project and check Coordination, then Schedule.	perm-folder-1 (Read Only) has the Rename button disabled. perm-folder-2 (Full Access) can be renamed and the new name persists. In the schedule, task 1 (No Access) is not displayed, task 2 (Limited Access) is visible and its % Complete can be edited to 55 and persists.
8	Log back in as administrator and delete the MJ team.	The team MJ is removed from the Teams panel.
9	Clear all folders in Coordination (cleanup).	All Coordination folders are deleted.
10	Delete the schedule created for this run (cleanup).	The schedule no longer exists in the schedule list.

4.3 Coordination

3 test cases | 50 steps

Covers the document management surface (upload, versioning, templates, sharing, rename and delete) and the ToDo / Workflow kanban board including its side panel, smart filters and exports.

COR-01 - Coordination - Files (upload, versioning, share, rename, delete)

Spec file	cypress/e2e/PTB_coordination/Files.cy.js
Suite name	Coordination - Files
Executed as	Test user (PM)
Pre-conditions	Project "Automation Project 3" open in the Coordination workspace. Fixture file content is reset before and after the run.
Step count	32

Step	Action	Expected Result
1	Upload a single file.	The file appears in the files list.
2	Upload three files in one action.	TestFile1, TestFile2 and TestFile3 are all listed.
3	Open one of the uploaded files.	The file viewer opens and displays the document.
4	Close the file tab and return to the files list.	The files list is displayed again.
5	Modify the source file content and re-upload the same file name.	The upload succeeds and the file is still listed once (a new version is created).
6	Open the file and select version V 1 from the version selector, then go back.	Version V 1 is selectable and its content is rendered.
7	Upload a folder containing sub-folders and files.	The folder upload-test-folder appears in the list.
8	Open the uploaded folder and inspect its contents, then navigate back to root.	SubFolder1, SubFolder2 and FolderFile1 are all present inside the folder.
9	Import a folder-structure template (search Test_MJ2, expand, select all folders, Import) and reload.	The template folder structure is imported into the project.
10	Select a file and click Download.	The download request is accepted and a download link is issued by email.
11	Create a new folder.	The new folder appears in the files list.
12	Open the new folder and upload a folder with files into it.	upload-test-folder is created inside the new folder.
13	Upload three separate files into the new folder.	The upload completes without error.
14	Verify the contents of the new folder, then navigate back to root.	upload-test-folder, TestFile1, TestFile2 and TestFile3 are all present.
15	Click Share and choose the Only Files scope, then Next.	The share wizard advances to the document selection step.
16	Select a file in the document tree and click Next.	The selected file is carried into the share settings step.
17	Enter a share title and password, set an expiration date (today + 5 days) and enable Include Download Link, then Next.	All share settings are accepted.
18	Open the Get Shareable Link tab and click Next.	The Review step is displayed.
19	Verify the share review summary.	Password and expiration date shown in the review match what was entered.

Step	Action	Expected Result
20	Open the Documents tab of the review.	The selected document is listed in the review.
21	Click Share via PTB.	The share is created successfully.
22	Dismiss any residual modal.	The screen returns to the files list.
23	Click Share again and choose the Folders & Files scope, then Next.	The wizard advances to document selection.
24	Expand the document tree, select a folder and click Next.	The selected folder is carried into the share settings step.
25	Enter share title, password, expiration date and enable Include Download Link, then Next.	All share settings are accepted.
26	Open the Get Shareable Link tab and click Next.	The Review step is displayed.
27	Verify the Folders & Files share review, check the Documents tab and click Share via PTB.	Review details match the entered settings and the share is created.
28	Dismiss any residual modal.	The screen returns to the files list.
29	Select the uploaded folder and rename it.	The folder is listed under its new name.
30	Select an uploaded file and rename it.	The file is listed under its new name.
31	Delete all files one by one (cleanup).	No files remain in the list.
32	Delete all folders one by one (cleanup).	No folders remain in the list.

COR-02 - Coordination - ToDo Side Panel, Smart Filter & Export

Spec file	cypress/e2e/PTB_coordination/sidePanelFilterExport.cy.js
Suite name	Coordination - ToDo Side Panel, Smart Filter & Export
Executed as	Test user (PM)
Pre-conditions	Project "Automation Project" open in Coordination > Workflows.
Step count	9

Step	Action	Expected Result
1	Create a new ToDo with a title and an assigned user.	The ToDo is created.
2	Check the board for the new ToDo.	The card is visible in the Backlog column.
3	Open the Backlog card and rename the Title from the side panel.	The side panel shows the renamed title.
4	Add a comment on the ToDo.	The comment is saved and displayed in the comment list.
5	Upload an attachment to the ToDo.	The attachment uploads and is listed against the ToDo.
6	Apply the "Created by me" smart filter.	The board is filtered to ToDos created by the logged-in user.
7	Open a filtered card and inspect the Created By field.	Created By shows a human display name (not a user id or e-mail).
8	Export the board to Excel.	The Excel export request succeeds and the file downloads.
9	Export the board to PDF.	The PDF export request succeeds and the file downloads.

COR-03 - Coordination Workflows - ToDo status flow

Spec file	cypress/e2e/PTB_coordination/todoFlow.cy.js
Suite name	Coordination Workflows - ToDo status flow
Executed as	Test user (PM)
Pre-conditions	Project "Automation Project" open in the Coordination workspace.
Step count	9

Step	Action	Expected Result
1	Open the Workflows tab and create a ToDo with an estimate of 16 and a due date two days out.	The ToDo is created with the supplied estimate and due date.
2	Check the board for the new ToDo.	The card is visible in the Backlog column.
3	Check the swim-lane the card sits in.	The card appears under the assigned user's row.
4	Drag the card from Backlog to Ready.	The card moves and is now in the Ready column.
5	Open the card and change the status to In Progress from the side panel.	The card moves to the In Progress column.
6	Open the ToDo detail page directly, set the status to On Hold, return to the board and refresh.	The card is shown in the On Hold column.
7	Reassign the ToDo to a different user.	The card's assignee id changes from the previous value.
8	Enable the Completed column on the board.	The Completed column becomes visible.
9	Set the status to Completed.	The card moves to the Completed column.

4.4 Schedule / Planning

13 test cases | 152 steps

The largest area of coverage. Includes task creation and hierarchy management, the full status-derivation matrix (Blank, OverDue, Delayed, WIP, Completed, Hold) at parent, child and grandchild level, side-panel editing, predecessor links, cash-flow finance entry with dashboard roll-up, Planned vs Actual baselining, smart filters, resource assignment with timesheets, and shareable links.

SCH-01 - Task Creation & Deletion

Spec file	cypress/e2e/PTB_Schedule/PTB_ScheduleTaskCreation/TasksCreationDeletion.cy.js
Suite name	Schedule - Task Creation & Deletion
Executed as	Test user (PM)
Pre-conditions	Project "Automation Project" open in the Planning workspace.
Step count	13

Step	Action	Expected Result
1	Create a new schedule.	The schedule is created and its name is visible.
2	Create the first task.	"New Task" exists in the grid.
3	Right-click the task and choose Add Above.	A task is inserted above; the task count increases.
4	Right-click the task and choose Add Below.	A task is inserted below; the task count increases.
5	Select the task, open the + menu and choose Add Child.	A child task is created under the selected task.
6	Select New Task - 2, open the + menu and choose Add Child.	A child task is created under New Task - 2.
7	Right-click the task and choose Add Child.	A second child task is created.
8	Select the child task, open the + menu and choose Add Below.	A sibling is added below the child task.
9	Delete a child task from the side panel.	The task is removed and the task count drops.
10	Select New Task - 2 and delete it from the side panel.	New Task - 2 no longer exists in the grid.
11	Delete the remaining child task via the right-click context menu.	The task is removed and the parent has no children left.
12	Double-click the task name and rename it to "Facade".	The grid shows the task under its new name.
13	Return to the schedule list and delete the schedule (cleanup).	The schedule no longer exists in the list.

SCH-02 - Task Status Verification (single task)

Spec file	cypress/e2e/PTB_Schedule/PTB_ScheduleStatusTCs/Schedule_Status.cy.js
Suite name	Schedule - Task Status Verification
Executed as	Test user (PM)
Pre-conditions	Project "Automation Project" open in the Planning workspace.
Step count	10

Step	Action	Expected Result
1	Create a new schedule.	The schedule is created and visible.
2	Create a task in the schedule.	"New Task" exists in the grid.
3	Change the start date to tomorrow.	Status resolves to BLANK.
4	Change the start date to 10 days before today.	Status resolves to OVERDUE.
5	Set the duration to 30 days.	Status resolves to DELAYED.
6	Move the end date 2 days earlier.	Status stays DELAYED and the duration recalculates to the expected derived value.
7	Set % Complete to 50.	Status resolves to WIP.
8	Set % Complete to 100.	Status resolves to COMPLETED.
9	Tick the On Hold checkbox.	Status resolves to HOLD.
10	Return to the schedule list and delete the schedule (cleanup).	The schedule no longer exists.

SCH-03 - Parent & Child Task Status Verification

Spec file	cypress/e2e/PTB_Schedule/PTB_ScheduleStatusTCs/ScheduleStatusChild.cy.js
Suite name	Schedule - Parent & Child Task Status Verification
Executed as	Test user (PM)
Pre-conditions	Project "Automation Project" open in the Planning workspace.
Step count	10

Step	Action	Expected Result
1	Create a new schedule.	The schedule is created and visible.
2	Create a parent task and add one child task to it.	Two rows exist: the parent (summary) and its child.
3	Select the child and set its start date to tomorrow.	Both parent and child show BLANK.
4	Set the child's start date to 10 days before today.	Both parent and child show OVERDUE.
5	Set the child's duration to 30 days.	Both parent and child show DELAYED.
6	Move the child's end date 2 days earlier.	Both rows stay DELAYED and both parent and child durations match the expected derived value.
7	Set the child's % Complete to 50.	Both parent and child show WIP.
8	Set the child's % Complete to 100.	Both parent and child show COMPLETED.
9	Tick On Hold on the child.	Both parent and child show HOLD.
10	Return to the schedule list and delete the schedule (cleanup).	The schedule no longer exists.

SCH-04 - Status roll-up with two child tasks

Spec file	cypress/e2e/PTB_Schedule/PTB_ScheduleStatusTCs/Schedule_2ChildStatus.cy.js
Suite name	Status Test cases for 2 and 3 childs
Executed as	Test user (PM)
Pre-conditions	Project "Automation Project" open in the Planning workspace.
Step count	9

Step	Action	Expected Result
1	Create a new schedule using the current date.	The schedule is created and visible.
2	Create a parent task.	"New Task" exists in the grid.
3	Add child 1 and child 2 under the parent, then set child 1's start date to tomorrow.	Three rows exist and child 1's status is BLANK.
4	Set child 2's start date to tomorrow and its % Complete to 16.	Parent = WIP, child 1 = BLANK, child 2 = WIP.
5	Set child 1's % Complete to 100.	Child 1 = COMPLETED and the parent's % Complete averages to 58.
6	Tick On Hold on child 1.	Parent = HOLD, child 1 = HOLD, child 2 = WIP.
7	Untick On Hold on child 1.	Parent returns to WIP and child 1 returns to COMPLETED.
8	Set child 2's duration to 10 days.	The parent's duration rolls up to 10 days.
9	Delete child 1.	The parent's status resolves to WIP based on the remaining child.

SCH-05 - Status and hierarchy with three child tasks

Spec file	cypress/e2e/PTB_Schedule/PTB_ScheduleStatusTCs/Schedule_3ChildStatus.cy.js
Suite name	Schedule - 3 Child Status Verification
Executed as	Test user (PM)
Pre-conditions	Project "Automation Project" open in the Planning workspace; the schedule is created in the setup hook.
Step count	6

Step	Action	Expected Result
1	Create a parent task with child 1, child 2 and child 3.	Four rows exist in the grid.
2	Move child 1's end date 2 days out.	The parent's duration equals the expected derived value (with weekend snap-back applied) and the parent's end date equals child 1's end date.
3	Set child 3's start date to 3 days before today.	Child 3 is OVERDUE and the delayed-days value matches the working days between its end date and today.
4	Set child 3's end date to today + 4, then right-click the parent and add a task below it.	Child 3 becomes DELAYED and a new task is added below the parent.
5	Indent the new task under the parent, then outdent it again.	The parent shows 4 indented children after indent and 3 after outdent.
6	Delete the schedule (cleanup).	The schedule no longer exists.

SCH-06 - Child-of-Child (grandchild) Task Status Verification

Spec file	cypress/e2e/PTB_Schedule/PTB_ScheduleStatusTCs/Schedule_CC_Status.cy.js
Suite name	Schedule - Child of Child Task Status Verification
Executed as	Test user (PM)
Pre-conditions	Project "Automation Project" open in the Planning workspace.
Step count	10

Step	Action	Expected Result
1	Create a new schedule.	The schedule is created and visible.
2	Create a parent task, add a child, then add a child to that child.	Three nested rows exist in the grid.
3	Set the grandchild's start date to tomorrow.	All three rows show BLANK.
4	Set the grandchild's start date to 10 days before today.	All three rows show OVERDUE.
5	Set the grandchild's duration to 30 days.	All three rows show DELAYED.
6	Move the grandchild's end date 2 days earlier.	All three rows stay DELAYED and all three durations match the expected derived value.
7	Set the grandchild's % Complete to 50.	All three rows show WIP.
8	Set the grandchild's % Complete to 100.	All three rows show COMPLETED.
9	Tick On Hold on the grandchild.	All three rows show HOLD.
10	Return to the schedule list and delete the schedule (cleanup).	The schedule no longer exists.

SCH-07 - Side Panel Task Editing

Spec file	cypress/e2e/PTB_Schedule/PTB_ScheduleSidePanel/ScheduleSidePanel.cy.js
Suite name	Schedule - Side Panel Task Editing
Executed as	Test user (PM)
Pre-conditions	Project "Automation Project" open in the Planning workspace.
Step count	13

Step	Action	Expected Result
1	Create a new schedule.	The schedule is created and visible.
2	Create a task in the schedule.	"New Task" exists in the grid.
3	Unhide any hidden Gantt columns.	All grid columns are visible for editing and assertion.
4	Select the task, open the side panel and rename it to "Test Rename".	The tree view shows the new task name.
5	Change the duration from 1D to 3D in the side panel.	The tree view shows a duration of 3 days.
6	Change the task Mode from Automatic to Manual.	The tree view shows mode "M" and both Constraint Type and Constraint Date are disabled.
7	Change the task Mode back from Manual to Automatic.	The tree view shows mode "A".

Step	Action	Expected Result
8	Set % Complete to 50 in the side panel, then close the panel.	The selected row shows 50% and the status column resolves to WIP.
9	Open the side panel and change the Start Date to the next working day with a time component.	The tree view shows the new start date (snapped to the next working day).
10	Change the End Date to start date + 1 working day at 17:00.	The tree view shows the new end date.
11	Open the Constraint Type dropdown and check the available options, then select "Finish No Later Than".	All eight constraint types are listed (As Soon As Possible, Must Start On, Start No Later Than, As Late As Possible, Must Finish On, Finish No Earlier Than, Finish No Later Than, Start No Earlier Than) and the tree view shows FNLT.
12	Change the Constraint Date.	The tree view shows the new constraint date (snapped to the previous working day).
13	Return to the schedule list and delete the schedule (cleanup).	The schedule no longer exists.

SCH-08 - Predecessor Links from the Side Panel

Spec file	cypress/e2e/PTB_Schedule/PTB_ScheduleSidePanel/Schedule_PredecessorSP.cy.js
Suite name	Schedule - Predecessor Side Panel
Executed as	Test user (PM)
Pre-conditions	Project "Automation Project" open in the Planning workspace.
Step count	6

Step	Action	Expected Result
1	Create a new schedule.	The schedule is created and visible.
2	Create 5 tasks and add a child task to three of them.	8 rows exist in total, of which 3 are indented child rows.
3	Select child 1 and add two predecessors from the side panel: task ID 3 as Finish-Start and task ID 4 as Start-Start.	Both predecessor links are saved and the status cell of the selected row renders correctly.
4	Select task ID 4 and add predecessor ID 5 with a 4-day lag and a Finish-Finish relationship.	The predecessor is saved with the lag applied and the status cell renders correctly.
5	Select task ID 7 and add predecessor ID 8 with a Start-Finish relationship.	The predecessor is saved and the status cell renders correctly.
6	Return to the schedule list and delete the schedule (cleanup).	The schedule no longer exists.

SCH-09 - Cash Flow - Booking / Invoicing, Nalco and Dashboard roll-up

Spec file	cypress/e2e/PTB_Schedule/PTB_ScheduleSidePanel/Schedule_CashFlow.cy.js
Suite name	Cash Flow - Booking/Invoicing + Nalco end-to-end
Executed as	Project administrator / PM
Pre-conditions	Project "Automation Project" open in the Planning workspace. Finance figures are entered in INR.
Step count	11

Step	Action	Expected Result
1	Create a new schedule.	The schedule is created and visible.
2	Create six tasks and rename them to Lot 1, Planning, Lot 2, Implementation, Lot 3 and Maintainece, then indent each child under its preceding lot.	Three summary tasks each with one child are shown, all names correct.
3	Open Lot 1 > side panel > Finances > Booking/Invoicing and add 1 booking and 3 invoices.	Booking section total = 10,00,000, invoice section total = 10,00,000 and the balance row shows 0.
4	Open the Nalco sub-tab for Lot 1 and add a Nalco Invoice and a Nalco Collection.	Nalco invoice total = 5,00,000 and Nalco collection total = 2,50,000.
5	Open Lot 2 > Finances > Booking/Invoicing and add 1 booking and 4 invoices.	Booking section total = 25,00,000, invoice section total = 25,00,000 and the balance row shows 0.
6	Open the Nalco sub-tab for Lot 2 and add a Nalco Invoice and a Nalco Collection.	Nalco invoice total = 10,00,000 and Nalco collection total = 5,00,000.
7	Reload the schedule and re-open the finances of both lots.	All booking, invoice, balance, Nalco invoice and Nalco collection values persist unchanged after reload.
8	Go back to the schedule list and mark the schedule as primary.	The schedule is flagged as the project's primary schedule.
9	Open the Project workspace Dashboard and check the finance cards.	Current Month Booking = 35,00,000, Current Month Invoice = 15,00,000, Nalco Invoice = 15,00,000 and Nalco Collection = 7,50,000.
10	Check the Financial Details per-lot summary on the dashboard.	Lot 1 and Lot 2 are listed with their booking and invoice values; Lot 3 (which has no financial values) is not listed.
11	Return to the Planning workspace and delete the schedule via multi-select (cleanup).	The schedule no longer exists in the list.

SCH-10 - Planned vs Actual (snapshot baselining and comparison)

Spec file	cypress/e2e/PTB_Schedule/PTB_PlannedVsActual/plannedvsactual.cy.js
Suite name	Planned vs Actual - Complete Test
Executed as	Test user (PM)
Pre-conditions	Project "Automation Project" open in the Planning workspace.
Step count	30

Step	Action	Expected Result
1	Create a new schedule.	The schedule is created and visible.
2	Add the first three standalone tasks (Task 1, Task 2, Task 3) using the context menu.	All three tasks exist in the grid.

Step	Action	Expected Result
3	Add Task 4 below Task 3, then add Task 5 as a child of Task 4.	Both tasks are created with the correct hierarchy.
4	Add Task 6 below Task 4, then add Task 7 as a child of Task 6.	Both tasks are created with the correct hierarchy.
5	Add Task 8 below Task 6.	8 tasks exist in total.
6	Create a snapshot with a name and description.	The snapshot is created and becomes the planned baseline.
7	Close the snapshot popup and unhide all Gantt columns.	All columns are visible for editing.
8	Task 1 - extend the end date forward by 5 days.	The end date is updated.
9	Task 2 - move the start date one working day earlier.	Status resolves to OVERDUE.
10	Task 3 - move the start date one working day earlier and set the duration to 2.	Status resolves to DELAYED.
11	Task 5 (child of Task 4) - set % Complete to 100.	Status resolves to COMPLETED.
12	Task 7 (child of Task 6) - set % Complete to 25.	Status resolves to WIP.
13	Task 8 - tick On Hold.	Status resolves to HOLD.
14	Open Planned vs Actual, tick Show Differences and click Compare.	The comparison view opens in differences-only mode.
15	Check which tasks are listed.	Only the 3 tasks with date or duration changes are listed, in order: Task 1, Task 2, Task 3.
16	Compare Planned Start against Actual Start for each listed task.	Task 1 start is unchanged; Task 2 and Task 3 starts differ from the baseline.
17	Compare Planned End against Actual End for each listed task.	Task 1 and Task 2 ends differ from the baseline; Task 3's end is unchanged.
18	Compare Planned Duration against Actual Duration for each listed task.	Task 1 and Task 3 durations differ from the baseline; Task 2's duration is unchanged.
19	Check the delta column for each listed task.	A non-empty delta is shown for all three changed tasks.
20	Close the comparison.	The comparison view closes and the Gantt is shown.
21	Re-open Planned vs Actual, untick Show Differences and click Compare.	The comparison view opens in full mode.
22	Check the number of rows listed.	All 8 tasks are listed.
23	Check the status column for each edited task.	Statuses read OVERDUE, DELAYED, COMPLETED, COMPLETED, WIP, WIP and HOLD in row order 1 to 7.
24	Check the percentage column for the edited tasks.	Rows 3 and 4 show 100% and rows 5 and 6 show 25%.
25	Apply the Delayed filter inside the comparison.	Exactly 1 row is shown - Task 3 with status DELAYED.
26	Apply the OverDue filter.	Exactly 1 row is shown - Task 2 with status OVERDUE.
27	Apply the WIP filter.	Exactly 2 rows are shown, both with status WIP.
28	Apply the Hold filter.	Exactly 1 row is shown - Task 8 with status HOLD.
29	Apply the Completed Task filter.	Exactly 2 rows are shown, both with status COMPLETED.
30	Close the comparison, return to the schedule list and delete the schedule (cleanup).	The schedule no longer exists.

SCH-11 - Smart Filters

Spec file	cypress/e2e/PTB_Schedule/PTB_FilterTCs/smartFilter.cy.js
Suite name	Schedule - Smart Filters
Executed as	Test user (PM)
Pre-conditions	Project "Automation Project" open in the Planning workspace.
Step count	11

Step	Action	Expected Result
1	Create a new schedule.	The schedule is created and visible.
2	Create 10 tasks with the current start date and unhide all columns.	10 tasks are present in the grid.
3	Set Task 1's % Complete to 10.	Status resolves to WIP.
4	Set Task 2's % Complete to 10.	Status resolves to WIP.
5	Tick On Hold for Task 3 and Task 4.	Both tasks move to the Hold status.
6	Set Task 5's and Task 6's % Complete to 100.	Both tasks resolve to COMPLETED.
7	Move Task 7's and Task 8's start dates back by 4 days.	Both tasks resolve to OVERDUE.
8	Move Task 9's and Task 10's start dates back 5 days and end dates forward 5 days.	Both tasks resolve to DELAYED.
9	Tick the Public checkbox for Task 1 and Task 2.	Both tasks are flagged as public.
10	Apply each smart filter in turn - WIP, Hold, Completed Task, OverDue, Delayed and Public - removing the filter after each check.	Each filter returns exactly the tasks that carry the corresponding status (and the correct pair of tasks for the Public filter), and the filter chip can be removed cleanly each time.
11	Return to the schedule list and delete the schedule (cleanup).	The schedule no longer exists.

SCH-12 - Resources and Timesheet (multi-user)

Spec file	cypress/e2e/PTB_Schedule/PTB_ResourcesTimesheet/ResourcesAndTimesheet.cy.js
Suite name	Schedule - Resources and Timesheet
Executed as	PM for setup and cleanup; non-PM resource (User MJ) for verification
Pre-conditions	Project "Automation Project" open in the Planning workspace.
Step count	12

Step	Action	Expected Result
1	Create a new schedule as PM.	The schedule is created and visible.
2	Create 5 tasks.	5 tasks are present in the grid.
3	Assign the resource User MJ to Task 1 via the right-click context menu.	The resource is assigned to Task 1.
4	Log out and log in as the non-PM user (User MJ).	The non-PM session is established.
5	Open the project and navigate to the schedule as the non-PM user.	The schedule opens for the non-PM user.
6	Check the left toolbar for the Delete action.	No Delete button is available to the non-PM user.

Step	Action	Expected Result
7	Check which tasks are visible.	Only the single assigned task is visible - not all 5.
8	Select the assigned task and open the side panel.	The side panel opens for the assigned task.
9	Open the Timesheet tab and add a timesheet entry of 5 hours.	The timesheet entry is saved.
10	Open Resource Usage and expand the resource row.	ACTUAL WORK for User MJ matches the 5 hours logged in the timesheet.
11	Log back in as PM and reopen the project.	The PM session is restored.
12	Delete the schedule (cleanup).	The schedule is removed.

SCH-13 - Share - Shareable Link (scope, settings and filters)

Spec file	cypress/e2e/PTB_Schedule/PTB_ShareSchedule/Sharablelink.cy.js
Suite name	Schedule - Share (Sharable Link)
Executed as	Test user (PM); shared link opened as an anonymous viewer
Pre-conditions	Project "Automation Project" open in the Planning workspace.
Step count	11

Step	Action	Expected Result
1	Create a schedule with 16 tasks and unhide all columns.	16 tasks are present in the grid.
2	Configure the task statuses: tasks 1-2 Delayed, 3-4 OverDue, 5-6 Blank, 7-8 WIP, 9-10 Completed, 11-12 OverDue, 13-14 Hold and 15-16 OverDue.	Each task resolves to its intended status.
3	Tick the Public checkbox for tasks 2, 4, 6, 8, 10, 12, 13, 14, 15 and 16.	All ten tasks are flagged public.
4	Convert tasks 11 and 12 to milestones and indent task 16 under task 15.	Two milestones exist, one indented child row is present and the total task count is still 16.
5	Open the Share modal with the All scope, copy the link and open it.	All 16 tasks are visible in the shared view.
6	Configure Link Settings - password, expiry (today + 30 days) and resource view - then open the link with the password.	The link prompts for the password, accepts it and all 16 tasks are visible.
7	Change task 16's % Complete to 20 and re-open the shared link.	The shared view shows task 16 at 20%.
8	Share again using the Public scope and open the link.	Only the ten Public-ticked tasks are visible.
9	Apply the Delayed filter, share with the All scope and open the link.	Only the two Delayed tasks are visible.
10	Re-apply the Delayed filter, share with the Public scope and open the link.	Only the one task that is both Delayed and Public is visible.
11	Return to the schedule list and delete the schedule (cleanup).	The schedule no longer exists.

4.5 Element Management System (EMS)

4 test cases | 54 steps

Covers the four EMS sub-modules - Model, System, Drawing and Batch. Each suite creates its own disposable building structure, exercises the module and deletes the structure at the end, so runs are self-contained and repeatable.

EMS-01 - Building Models - Model

Spec file	cypress/e2e/PTB_EMS/model.cy.js
Suite name	EMS - Building Models - Model
Executed as	Test user (PM)
Pre-conditions	Project "Automation Project 3". The setup hook guarantees at least one Coordination folder exists, because Step 11's folder picker only lists child folders under Root.
Step count	15

Step	Action	Expected Result
1	Create a Building Structure (EMS).	The building structure is created and opened.
2	Add a Group via the + menu.	A new group node appears in the tree.
3	Rename the group to AutGrp_01 and check its node properties.	The tree shows the new name, node status is "Unassigned" and the Derived Status control is enabled.
4	Right-click the group and add an Element.	A child element node is created under the group.
5	Open the element and switch to the Status Tracker tab.	The Status Tracker tab is displayed for the element.
6	Add two Manual Statuses.	Two manual status rows are present.
7	Set Manual Status 1 to "Assigned".	The status value is saved as Assigned.
8	Delete Manual Status 2.	Manual Status 2 is no longer listed.
9	Open the To Dos tab and create a To Do with a title, assigned user and today's planned date.	The To Do is created and appears in the element's To Do list.
10	Open the Documents tab and upload a file.	The uploaded document is listed against the element.
11	Add a Project Files / Folders entry using the "Select folders" picker.	The attached project folder is displayed in the Documents tab.
12	Open the Cost tab, add a cost with Quantity = 100 and Cost/Unit = 10.	Cost (USD) is calculated as Quantity x Cost/Unit = 1000.
13	Open the Tickets tab, create a new ticket with a title, assignee, category and type, attach a file and click Create.	The ticket is created and listed against the element.
14	Open the created ticket.	The application navigates to the Tickets module showing that ticket.
15	Return to the EMS home screen and delete the building structure (cleanup).	The building structure is removed.

EMS-02 - Building Models - System

Spec file	cypress/e2e/PTB_EMS/system.cy.js
Suite name	EMS - Building Models - System
Executed as	Test user (PM)
Pre-conditions	Project "Automation Project 3". A dedicated building structure is created in the setup hook. System names, vent types and element prefixes are randomised from a Schuco keyword library.
Step count	12

Step	Action	Expected Result
1	Create a System.	The system node is created and selected in the tree.
2	Right-click the system and add a Sub-System.	A sub-system child node is created under the system.
3	Open the sub-system's General tab and fill the System Subtype Name, Vent Type and dimension fields.	The values are saved and the sub-system node in the tree is renamed to the subtype name.
4	Open the Articles tab and add an Article.	At least one article row is present.
5	Edit the article: set the name, packaging unit, UOM type, quantity, price and package unit size on screen 1, then fill all 28 property fields on screen 2 and click Done.	All general-information and property values are accepted and saved.
6	Export the articles to Excel.	The export file downloads and contains the article name that was entered.
7	Open the Cost tab and add a cost with Quantity = 10 and Cost/Unit = 100.	Cost (USD) is calculated as Quantity x Cost/Unit = 1000.
8	Open the Tracking Templates tab and add a tracking template.	The selected tracking template is listed against the sub-system.
9	Open the Elements tab and create 10 elements using the prefix / count / start / length generator.	10 elements are created.
10	Switch to the Model tab.	All 10 generated elements are fetched and shown under the model tree.
11	Return to the System module, open the sub-system's Document tab and upload a file.	The uploaded document is listed against the sub-system.
12	Delete the building structure (cleanup).	The building structure is removed.

EMS-03 - Building Models - Drawing

Spec file	cypress/e2e/PTB_EMS/drawing.cy.js
Suite name	EMS - Building Models - Drawing
Executed as	Test user (PM)
Pre-conditions	Project "Automation Project 3". A dedicated building structure is created in the setup hook. The fixture is a real architectural PDF (West-Elevation.pdf) that renders taggable objects.
Step count	11

Step	Action	Expected Result
1	Open the Drawing module.	The Drawing module is displayed.
2	Add a folder.	A new folder node appears in the drawing tree.
3	Upload a drawing (PDF) into the folder.	The drawing is uploaded into the selected folder.

Step	Action	Expected Result
4	Open the drawing.	The PDF viewer renders the drawing to canvas.
5	Switch to the Model module and create a Group and an Element.	The group is created, renamed, and has one child element.
6	Open the element's General tab and set the Node Name to a Schuco element name.	The element node is renamed in the tree.
7	Open the Tags tab and add a tag on the drawing.	The tag is created and shown on the drawing.
8	Return to the Drawing module and open the Annotation History.	Annotation History lists at least one row reflecting the tag that was added, and the history saves successfully.
9	Open the Downloads panel and choose "Download With Annotations", then confirm the page-range modal.	The download API is called and returns a success status (200, 201 or 206).
10	Open the Downloads panel and choose "PDF Download".	The PDF download API is called and returns HTTP 200.
11	Delete the building structure (cleanup).	The building structure is removed.

EMS-04 - Building Models - Batch and QR Code

Spec file	cypress/e2e/PTB_EMS/batch.cy.js
Suite name	EMS - Building Models - Batch
Executed as	Test user (PM)
Pre-conditions	Project "Automation Project 3". The setup hook creates a disposable building structure with one group containing 5 elements.
Step count	16

Step	Action	Expected Result
1	Navigate to the Batch module and create a new Batch using the + button.	The new batch is created and listed.
2	Open the new batch and switch to the Elements tab.	The Elements tab of the batch is displayed.
3	Click Attach Elements.	The "Attach EMS Nodes" modal opens.
4	Select all elements in the modal and confirm.	5 element rows are attached and visible in the batch.
5	Set the first element's BatchStatus to "In Progress".	The element's batch status is updated.
6	Go to the Tracking page, confirm the Model tab is active, select one element and click Import Template.	The template selection dialog opens.
7	Pick a template and save.	The template is attached to the selected element and shown on the tracking row.
8	Select all elements using the header checkbox.	All tracking rows are selected.
9	Open the QR Code screen via the QR scanner icon.	The QR Code screen opens.
10	Pick a template in the QR sidebar, name it, save and click Save to Project.	The QR template is saved to the project.
11	Click the Print QR icon.	The application opens a print-preview window (window.open is invoked).
12	Close the QR Code screen using the X icon.	The QR Code screen closes and the tracking page is shown.

Step	Action	Expected Result
13	Switch to the Batch tab, select the batch and open the QR Code screen.	The QR Code screen opens in batch context.
14	Pick a template in the QR sidebar, name it and save (batch context).	The QR template is saved for the batch.
15	Click the Print QR icon in batch context, then close the popup.	A print-preview window is opened and the popup closes cleanly.
16	Delete the building structure (cleanup).	The building structure is removed.

4.6 Field Logs

1 test case | 8 steps

Covers the Form Template builder and the full lifecycle of a field log built from that template - creation, distribution, sending, reopening, PDF export and deletion.

FL-01 - Form Template create, send, reopen, export & delete

Spec file	cypress/e2e/PTB_FieldLogs/logCreatDel.cy.js
Suite name	Field Logs - Form Template create, send, reopen, export & delete
Executed as	Test user (PM)
Pre-conditions	Project "Automation Project" is open. A uniquely named template is created per run so the dropdown assertion is unambiguous.
Step count	8

Step	Action	Expected Result
1	Create a Form Template containing a Layout, a Checklist and an Image, tagged "Logs".	The template is created and the name field holds exactly the value entered.
2	Switch to the Field Operations workspace and open the Logs tab.	The Logs tab is displayed.
3	Create a new log and pick the template created in step 1 from "Use Template", then confirm.	The log is created from the selected template.
4	Add a user to the Distribution List.	The user is added to the distribution list.
5	Click Send and submit.	The confirmation "The email was successfully sent." is displayed.
6	Reopen the log.	The log reopens and the Send action is enabled again.
7	Export the log to PDF.	The /logforms/.../pdf endpoint returns HTTP 200 and the file downloads.
8	Delete the log (cleanup).	The log is removed from the Logs list.

5. Notes and Current Limitations

The following points are relevant when reading the coverage above:

Disabled cases	Four cases in the Import Project Creation suite (PC-02: add a checklist row as PM, non-PM read-only access, edited-XLSX re-upload and row cleanup) remain in the code but are commented out. The affected controls in cmacs-compact-table only appear under real operating-system hover, which Cypress cannot reproduce reliably. These four scenarios are covered by manual testing.
Real-event dependency	Drag-and-drop (team membership, ToDo kanban cards) and some permission cells use Angular CDK, which ignores synthetic events. These interactions are driven through cypress-real-events, which requires the browser window to stay focused during the run.
Ordered execution	Test isolation is switched off deliberately. Each spec is a single continuous journey - a step depends on the state left by the step before it - which mirrors how the application is actually used, but means a spec should be run whole rather than by individual step.
Environment dependency	Suites run against the shared staging environment and use named projects (Automation Project, Automation Project 3) and named teams and templates. Renaming or deleting these in staging will break the affected suites.
Date-sensitive assertions	Status derivation (OverDue, Delayed, Blank) is calculated from today's date with working-day and weekend snapping applied. The helpers in cypress/support/utils/dateUtils.js derive the expected values rather than hard-coding them, so results stay stable on any weekday.
Not yet automated	Areas not represented in the current suite include Tickets as a standalone module, Dashboards beyond the finance cards exercised by SCH-09, Portal and Home, and company-level (as opposed to project-level) administration.

6. How to Execute

Interactive runner (pick and watch a spec):

```
npm run cy:open
```

Headless run of the whole suite, producing the Mochawesome HTML report:

```
npm run cy:run
```

Single suite, for example the Planned vs Actual coverage:

```
npx cypress run --spec "cypress/e2e/PTB_Schedule/PTB_PlannedVsActual/plannedvsactual.cy.js"
```

The report is written to cypress/reports with charts and screenshots of any failures embedded inline.